Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/23/2006
PI Name:	Driscoll, James Ph.D.		
Project Title:	A Flame Extinction Database for Unsteady Conditions Obtained Using a Flame-Vortex Experiment		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	48109-2118	**Congressional District:**	15
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03
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No. of PhD Candidates:		**No. of Master' Degrees:**	
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No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:			
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Performance Goal Text:			
	This is a fundamental science study which will add to NASA’s steady-state flame extinction database some much-needed new information: scaling laws for flame extinction limits under well characterized 'canonical' unsteady conditions. Previous NASA studies characterized the extinction of steady (counterflow) flames. However, it is unrealistic to think that the unsteady effects will not be important in most real combustion process. The goal is to produce submodels for unsteady flame extinction that adds an important new parameter – the residence time – in addition to the stretch rate. This realistic submodel of unsteady flame extinction is needed: as a submodel to turbulent combustion Large Eddy Simulations, such as the Meneveau Coherent Flamelet submodel as a submodel to fire codes, such as the code of Howard Baum at NIST, as a submodel to predict engine flameout in NASA’s propulsion systems. An ideal way to measure the unsteady flame extinction limits and add them to the existing steady-state database is to use the existing University of Michigan Flame Vortex Microgravity experiment. It provides a simple geometry, yet retains the physics		

Task Description:	of vortical motions in shear layers and turbulence, and removes the complications of buoyancy. We have proven that we can measure the appropriate stretch rates and residence times. DNS modelers around the world have chosen this flame-vortex data set as their 'test case'. We plan to: 1. Image the entire time history of the extinction process – including flame stretch rates (deduced from flame area measurements), flame geometry, flame curvature, 2. Image the time history of CH and OH chemiluminescence intensities during extinction, for comparison to simulations; 3. Measure the flame extinction stretch rate using flame area data and local PIV measurements, for various characteristic times that quantify the unsteadiness 4. Measure effects of residence time since extinction depends on residence time 5. Vary the following: oxygen/nitrogen ratio, suppressants added (CO2, N2, CF3Br), characteristic time scale and strength of the unsteady strain field, fuel type 6. Compare extinction limits to numerical simulations of Patniak and Kailasanath who have simulated our exact geometry with full chemistry.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this reporting period.
Bibliography Type:	Description: (Last Updated:)